

Day 15: Area of a Trapezoid

LESSON + PRACTICE

MathCraft CPA · Chandler Park Academy · Grade 6 Mathematics · Professor Xavier Honablu, M.Ed.

Standard: 6.G.A.1 — find the area of special quadrilaterals and polygons by composing into rectangles or decomposing into triangles and other shapes; MP.7, MP.8

I can: find the area of a trapezoid two ways — by decomposing it into a rectangle and triangles, and with the formula $A = \frac{1}{2}(b_1 + b_2)h$ — and find a missing base or height.

Homework: Earn 90% on IXL "Area of trapezoids" (GG, 6th grade) — [ixl.com/math/grade-6/area-of-a-trapezoid](https://www.ixl.com/math/grade-6/area-of-a-trapezoid)

Remember: show ALL work by hand in your math notebook (graph paper is best) and label every answer with units.

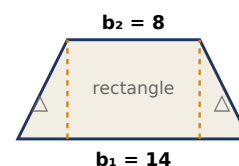
Learn

1 · A trapezoid has exactly one pair of parallel sides

Those two parallel sides are the **bases**, written b_1 ("b-sub-one") and b_2 ("b-sub-two"). They are usually different lengths — that is what makes a trapezoid lean. The **height** h is the perpendicular distance between the two bases, never a slanted side.

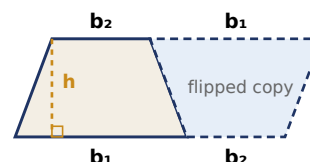
2 · Decompose: one rectangle + two triangles

Drop a height from each end of the short base. The trapezoid splits into a **rectangle** in the middle and a **triangle** on each end. Find the three areas and add — this always works, and it uses only Day 5 and Day 13 skills.



3 · Compose: two copies make a parallelogram

Make a copy of the trapezoid, spin it 180° , and slide it against the original. The two together form a **parallelogram** whose base is $b_1 + b_2$ and whose height is the same h . That parallelogram has area $(b_1 + b_2) \times h$, and one trapezoid is half of it: $A = \frac{1}{2} \times (b_1 + b_2) \times h$. Read it as: "add the two bases, multiply by the height, cut in half."



two copies = one parallelogram, base $b_1 + b_2$

4 · Backwards: find a missing base

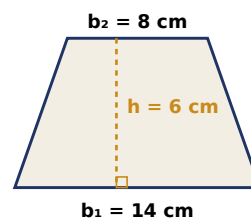
Double the area to undo the half: $2A = (b_1 + b_2) \times h$. Divide by h to get the sum of the bases, then subtract the base you know.

Worked Examples

Example 1. A trapezoid has bases 14 cm and 8 cm and height 6 cm. Find its area with the formula.

- Step 1: Add the bases: $14 + 8 = 22$.
- Step 2: Multiply by the height: $22 \times 6 = 132$.
- Step 3: Cut in half: $132 \div 2 = 66$.

✓ **Area = 66 sq cm**



Example 2. Find the area of that same trapezoid by decomposing it (rectangle + two triangles).

- Step 1: The middle rectangle is 8 by 6: $8 \times 6 = 48$.
- Step 2: The two triangles together have a base of $14 - 8 = 6$ and height 6.
- Step 3: Triangles together: $\frac{1}{2} \times 6 \times 6 = 18$.
- Step 4: $48 + 18 = 66$ — the same answer as the formula. ✓

✓ **Area = 66 sq cm**

Example 3. A trapezoid has an area of 72 sq in, one base of 10 in, and a height of 8 in. Find the other base.

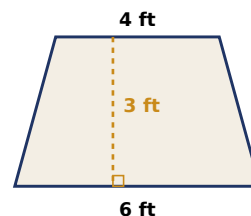
- Step 1: Double the area: $2 \times 72 = 144 = (b_1 + b_2) \times 8$.
- Step 2: Divide by the height: $144 \div 8 = 18$, so $b_1 + b_2 = 18$.
- Step 3: Subtract the known base: $18 - 10 = 8$.

✓ **The other base is 8 in**

Example 4. A tabletop is a trapezoid with bases 6 ft and 4 ft and height 3 ft. Laminate costs \$8 per square foot. What is the cost?

- Step 1: Area: $\frac{1}{2} \times (6 + 4) \times 3 = \frac{1}{2} \times 30 = 15$ sq ft.
- Step 2: Cost: $15 \times \$8 = \120 .

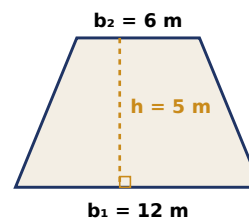
✓ **15 sq ft; \$120**



Practice — show your work in your notebook

1. Find the area of the trapezoid.

Answer: _____

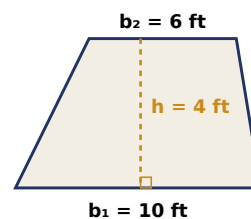


2. A trapezoid has bases 9 in and 7 in and a height of 4 in. Find its area.

Answer: _____

3. Distractor check. This trapezoid shows bases 10 ft and 6 ft, a height of 4 ft, and a slanted side of 5 ft. Find the area.

Answer: _____



4. Find the area of a trapezoid with bases 20 cm and 14 cm and height 9 cm.

Answer: _____

5. A trapezoid has an area of 60 sq ft, bases 8 ft and 7 ft. Find its height.

Answer: _____

6. A trapezoid has an area of 84 sq m, one base of 9 m, and a height of 8 m. Find the other base.

Answer: _____

7. Solve by decomposing (not the formula): a trapezoid has bases 16 in and 10 in and height 5 in. Show the rectangle and triangle areas, then the total.

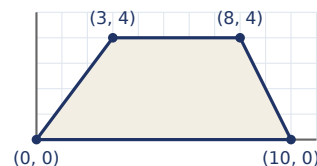
Answer: _____

8. Find the mistake. For a trapezoid with bases 11 cm and 5 cm and height 6 cm, Marcus wrote $(11 + 5) \times 6 = 96$ sq cm. What went wrong?

Answer: _____

9. Find the area of the trapezoid plotted on the grid.

Answer: _____



10. A garden is a trapezoid with bases 7.5 m and 4.5 m and a height of 6 m. Find its area.

Answer: _____

11. A window is a trapezoid with bases $3\frac{1}{2}$ ft and $2\frac{1}{2}$ ft and height 4 ft. Find its area.

Answer: _____

12. A deck is made of a rectangle 12 ft by 5 ft with a trapezoid attached: bases 12 ft and 6 ft, height 4 ft. Find the total area, then the cost of decking at \$9 per square foot.

Answer: _____

Answer Key & Worked Solutions

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1. Find the area of the trapezoid.

- Step 1: Add the bases: $12 + 6 = 18$.
- Step 2: $18 \times 5 = 90$.
- Step 3: $90 \div 2 = 45$.

✓ **Answer: 45 sq m**

2. A trapezoid has bases 9 in and 7 in and a height of 4 in. Find its area.

- Step 1: $9 + 7 = 16$.
- Step 2: $16 \times 4 = 64$.
- Step 3: $64 \div 2 = 32$.

✓ **Answer: 32 sq in**

3. Distractor check. This trapezoid shows bases 10 ft and 6 ft, a height of 4 ft, and a slanted side of 5 ft. Find the area.

- Step 1: Use the two parallel bases and the perpendicular height only.
- Step 2: $\frac{1}{2} \times (10 + 6) \times 4 = \frac{1}{2} \times 64 = 32$.
- Step 3: The 5 ft slanted side is a distractor.

✓ **Answer: 32 sq ft**

4. Find the area of a trapezoid with bases 20 cm and 14 cm and height 9 cm.

- Step 1: $20 + 14 = 34$.
- Step 2: $34 \times 9 = 306$.
- Step 3: $306 \div 2 = 153$.

✓ **Answer: 153 sq cm**

5. A trapezoid has an area of 60 sq ft, bases 8 ft and 7 ft. Find its height.

- Step 1: Double: $120 = (8 + 7) \times h = 15h$.
- Step 2: $h = 120 \div 15 = 8$.

✓ **Answer: 8 ft**

6. A trapezoid has an area of 84 sq m, one base of 9 m, and a height of 8 m. Find the other base.

- Step 1: Double: $168 = (9 + b_2) \times 8$.
- Step 2: $168 \div 8 = 21$, so $9 + b_2 = 21$.
- Step 3: $b_2 = 21 - 9 = 12$.

✓ **Answer: 12 m**

7. Solve by decomposing (not the formula): a trapezoid has bases 16 in and 10 in and height 5 in. Show the rectangle and triangle areas, then the total.

- Step 1: Middle rectangle: $10 \times 5 = 50$.
- Step 2: Two end triangles together: base $16 - 10 = 6$, height 5 $\rightarrow \frac{1}{2} \times 6 \times 5 = 15$.
- Step 3: Total: $50 + 15 = 65$.
- Step 4: Check with the formula: $\frac{1}{2} \times (16 + 10) \times 5 = 65$. ✓

✓ **Answer: 65 sq in**

8. Find the mistake. For a trapezoid with bases 11 cm and 5 cm and height 6 cm, Marcus wrote $(11 + 5) \times 6 = 96$ sq cm. What went wrong?

- Step 1: Marcus found the area of the whole doubled parallelogram.
- Step 2: One trapezoid is half of it: $96 \div 2 = 48$.
- Step 3: He forgot to cut it in half.

✓ **Answer: He left out the $\frac{1}{2}$; correct area is 48 sq cm**

9. Find the area of the trapezoid plotted on the grid.

- Step 1: Bottom base: from (0, 0) to (10, 0) = 10 units.
- Step 2: Top base: from (3, 4) to (8, 4) = 5 units.
- Step 3: Height: $4 - 0 = 4$ units.
- Step 4: $\frac{1}{2} \times (10 + 5) \times 4 = \frac{1}{2} \times 60 = 30$.

✓ **Answer: 30 square units**

10. A garden is a trapezoid with bases 7.5 m and 4.5 m and a height of 6 m. Find its area.

- Step 1: $7.5 + 4.5 = 12$.
- Step 2: $12 \times 6 = 72$.
- Step 3: $72 \div 2 = 36$.

✓ **Answer: 36 sq m**

11. A window is a trapezoid with bases $3\frac{1}{2}$ ft and $2\frac{1}{2}$ ft and height 4 ft. Find its area.

- Step 1: $3\frac{1}{2} + 2\frac{1}{2} = 6$.
- Step 2: $6 \times 4 = 24$.
- Step 3: $24 \div 2 = 12$.

✓ **Answer: 12 sq ft**

12. A deck is made of a rectangle 12 ft by 5 ft with a trapezoid attached: bases 12 ft and 6 ft, height 4 ft. Find the total area, then the cost of decking at \$9 per square foot.

- Step 1: Rectangle: $12 \times 5 = 60$ sq ft.
- Step 2: Trapezoid: $\frac{1}{2} \times (12 + 6) \times 4 = \frac{1}{2} \times 72 = 36$ sq ft.
- Step 3: Total: $60 + 36 = 96$ sq ft.
- Step 4: Cost: $96 \times \$9 = \864 .

✓ **Answer: 96 sq ft; \$864**